# Assignment

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# Examples of Sets

Even numbers:

$$A = \{x \in \mathbb{Z} \mid x \text{ is even}\}$$

Odd numbers:

$$B = \{x \in \mathbb{Z} \mid x \text{ is odd}\}$$

# Examples of Sets (contd.)

Natural numbers less than 10:

$$C = \{x \in \mathbb{N} \mid x < 10\}$$

Multiples of 5:

$$D = \{x \in \mathbb{Z} \mid x = 5n,\ n \in \mathbb{Z}\}$$

# Examples of Sets (contd.)

Prime numbers:

$$E = \{x \in \mathbb{N} \mid x \text{ is prime}\}$$

Integers between -5 and 5:

$$F = \{x \in \mathbb{Z} \mid -5 \leq x \leq 5\}$$

# Examples of Sets (contd.)

Perfect squares:

$$G = \{x \in \mathbb{N} \mid x = n^2,\ n \in \mathbb{N}\}$$

Real numbers greater than 0:

$$H = \{x \in \mathbb{R} \mid x > 0\}$$

# Examples of Sets (contd.)

Students scoring above 80:

$$I = \{x \in S \mid \text{score}(x) > 80\}$$

Letters in the word APPLE:

$$J = \{x \in \{A, B, C, \ldots, Z\} \mid x \text{ appears in "APPLE"}\}$$

# Inclusion-Exclusion Principle

General Formula:

$$\left|\bigcup_{i=1}^{\infty} A_i\right| = \sum_{k=1}^{\infty}(-1)^{k+1}\left(\sum_{1\le i_1<i_2<\cdots<i_k}\left|A_{i_1}\cap A_{i_2}\cap\cdots\cap A_{i_k}\right|\right)$$

# Proof Sketch

1. Finite Truncation:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \left|A_{i_1} \cap \cdots \cap A_{i_k}\right|$$

2. Limit Process: As $n \to \infty$, union approaches infinite union.

# Proof Sketch (contd.)

3. Element-wise Counting:

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m+1}\binom{m}{m} = 1$$

4. Conclusion: Each element counted exactly once.

# Types of Sets

| **Type of Set** | Example |
|---|---|
| **Numerical** | Sensor readings: $\{23.1, 24.5, 25.0\}$ |
| **Categorical** | Sentiment labels: $\{\text{Positive}, \text{Negative}\}$ |
| **Time Series** | Stock prices: $\{(t_1, 100), (t_2, 102)\}$ |
| **Ordinal** | Satisfaction: $\{\text{Poor}, \text{Fair}, \text{Good}\}$ |

# Types of Sets (contd.)

| **Type of Set** | Example |
|---|---|
| **Image Set** | MNIST digits $\{0, 1, 2, \ldots, 9\}$ |
| **Text Set** | NLP corpus: {"AI is powerful"} |
| **Web/API Set** | Tweets: $\{\text{Tweet}_1, \text{Tweet}_2\}$ |
| **Multivariate** | Patient records: $\{(\text{Age}, \text{BP})\}$ |

# Partially Ordered Sets (Posets)

Definition: A poset $(P, \leq)$ satisfies:

- Reflexive: $a \leq a$
- Antisymmetric: $a \leq b$ and $b \leq a \Rightarrow a = b$
- Transitive: $a \leq b, b \leq c \Rightarrow a \leq c$

Not all elements need to be comparable.

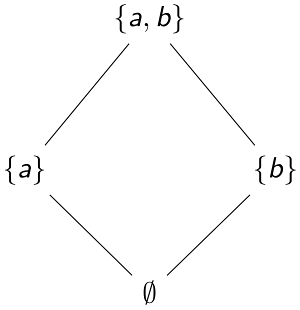
# Poset Example: Divisibility on $\{1, 2, 4\}$

$$1 \leq 2 \leq 4$$

$$\begin{tikzcd}
4 \arrow[d, no head] \\
2 \arrow[d, no head] \\
1
\end{tikzcd}$$

# Poset Example: Power set of $\{a, b\}$

$$\emptyset \subseteq \{a\}, \{b\} \subseteq \{a, b\}$$



# Poset Example: Divisibility on $\{1, 2, 3, 6\}$

$$1 \leq 2 \leq 6, \quad 1 \leq 3 \leq 6$$